

Technical Work Package

Hanoi 3D Building Data, Demolition Detection, and 3D Map Tile API Pilot

1. Project Context

We are building a 3D Building Data Platform and 3D Map Tile API for a production mapping platform in Vietnam.

The first pilot area is Hanoi.

Hanoi is undergoing active urban change. In several areas, old houses and buildings have been demolished to create or expand roads and transport infrastructure. Existing open or global datasets may still contain these previous buildings. This creates a serious map quality issue: if we directly render all buildings from Google-derived datasets, OSM, or the Munich/TUM GlobalBuildingAtlas dataset, the 3D map may show buildings that no longer exist.

Therefore, this project is not a pure 3D rendering task. It is a building data lifecycle and quality-control platform.

The pilot must identify, classify, and render buildings with confidence. It must distinguish between:

```
active_building  
demolished_candidate  
demolished_verified  
new_construction_candidate  
under_construction_candidate  
low_confidence_building  
do_not_render
```

The expected result is a district-focused Hanoi MVP that can ingest multiple building data sources, create a canonical building database, detect obsolete/demolished building candidates, generate confidence scores, produce LoD1 3D buildings, serve 3D Tiles and vector tiles, and provide a QA dashboard.

2. Pilot Geography

2.1 Primary City

Hanoi, Vietnam.

2.2 Operating AOI Unit

Use legacy Hanoi urban districts / Quận as the first operating AOI segmentation.

The vendor should support both:

```
legacy_district_name  
legacy_district_code  
current_ward_name  
current_ward_code
```

This is required because district names are still practical for project communication, map operations, and AOI planning, while current administrative operations in Vietnam now use the provincial/city level and commune/ward level.

2.3 Recommended First AOI

The first pilot should cover one to three dense Hanoi urban districts, selected by Tasco.

Suggested starting candidates:

```
Ba Dinh  
Hoan Kiem  
Dong Da  
Hai Ba Trung  
Cau Giay  
Thanh Xuan  
Hoang Mai  
Long Bien
```

Final AOI selection will depend on:

```
road-construction and demolition activity  
building density  
business priority  
availability of current imagery  
availability of field verification  
OSM/Google/Munich dataset disagreement  
expected demo value
```

2.4 Road-Corridor Priority

Within each selected district, the vendor should prioritize:

```
road-widening corridors
new road corridors
demolition zones
urban redevelopment corridors
major intersections
transport hubs
areas where old building datasets conflict with current road geometry
```

Tasco may provide road project boundaries, right-of-way polygons, or corridor buffers. If these are not available initially, the vendor should create a provisional detection method using road geometry, recent imagery, and building-source disagreement.

3. Core Problem to Solve

The main data problem is that previous building datasets may be stale.

A building may appear in:

```
Google building data
Google Open Buildings / Open Buildings Temporal
Munich/TUM GlobalBuildingAtlas
OSM building polygons
older satellite imagery
```

But the building may no longer exist because it has been demolished for road construction.

The platform must not blindly render all detected buildings. It must assign each building a lifecycle state, confidence score, and render policy.

4. Required Data Sources

The vendor should design the system to support the following data sources.

4.1 Required Sources for MVP

```
OSM building polygons
Google-derived building footprints or Google Open Buildings where available
Google Open Buildings 2.5D Temporal or equivalent temporal raster source
Munich/TUM GlobalBuildingAtlas building polygons, heights, and LoD1 data
Current road network
```

Current or recent satellite imagery, if available
Tasco-provided road construction / demolition corridors, if available

4.2 Optional Sources

Tasco field survey imagery
street-view or dashcam imagery
manual QA observations
government/cadastral data, if available and legally usable
Microsoft TEMPO temporal building density and height raster, if available and license-approved
commercial high-resolution satellite imagery

4.3 Data Source Classification

The system must treat sources differently.

Individual building polygon source:
OSM, Google building polygons, TUM/GlobalBuildingAtlas polygons

Individual height / LoD1 source:
OSM height, OSM building:levels, TUM/GlobalBuildingAtlas height and LoD1

Temporal raster prior:
Google Open Buildings 2.5D Temporal, TEMPO, other raster height/density datasets

Operational evidence:
road construction corridor, demolition polygon, field survey, QA notes

Raster sources must not be directly treated as individual-building truth. They should be used as temporal priors, freshness signals, and QA triggers.

5. Expected System Architecture

[Source Data]
OSM buildings
Google building data
Google Open Buildings Temporal
TUM/GlobalBuildingAtlas
Road network
Road construction corridors

Recent imagery
Field survey evidence

↓

[Ingestion Layer]
CRS normalization
source metadata capture
raw source snapshot storage
license metadata storage

↓

[Conflation Layer]
building matching
IoU comparison
centroid distance comparison
area ratio comparison
height source comparison
canonical building ID generation

↓

[Lifecycle and Change Detection Layer]
active building detection
demolished candidate detection
new construction candidate detection
road-corridor conflict detection
temporal raster disagreement detection

↓

[Confidence and QA Layer]
footprint confidence
height confidence
freshness confidence
demolition confidence
overall confidence
manual QA status

↓

[Canonical Building Store]
PostgreSQL/PostGIS
3DCityDB-compatible semantic model
source lineage
QA ledger

versioning

↓

[Tile Generation]

MVT vector tiles

LoD1 building extrusion

3D Tiles / GLB / B3DM

tileset.json generation

↓

[API Serving]

tile API

building metadata API

QA API

version API

diff API

↓

[Client and Dashboard]

MapLibre extrusion viewer

Cesium or TerriaJS 3D viewer

QA dashboard

source disagreement dashboard

demolition candidate dashboard

6. Canonical Building Data Model

The vendor must not store only geometry and height. Every building must have lineage, confidence, lifecycle state, and render policy.

6.1 Core Building Fields

```
tasco_building_id
geom_2d
geom_3d_lod1
height_m
min_height_m
legacy_district_name
legacy_district_code
current_ward_name
current_ward_code
```

```
source_lineage
footprint_confidence
height_confidence
freshness_confidence
demolition_confidence
overall_confidence
building_lifecycle_status
qa_status
render_policy
valid_from
valid_to
created_at
updated_at
```

6.2 Building Lifecycle Status

```
active_building
demolished_candidate
demolished_verified
new_construction_candidate
under_construction_candidate
low_confidence_building
unknown_status
```

6.3 Render Policy

```
render_production
render_qa_only
render_transparent_low_confidence
hide_from_production
do_not_render
```

6.4 Source Lineage Example

```
{
  "osm": {
    "osm_id": "way/123456",
    "version": "2026-xx",
    "has_height": false,
    "has_levels": true
  },
  "google": {
    "source_type": "building_polygon_or_temporal_prior",
    "confidence": 0.84,
```

```

    "source_year": "2023"
  },
  "tum_global_building_atlas": {
    "polygon_available": true,
    "height_m": 17.2,
    "lod1_available": true
  },
  "road_corridor": {
    "intersects_current_road_corridor": true,
    "corridor_source": "tasco_or_public"
  },
  "manual_qa": {
    "status": "not_reviewed"
  }
}

```

7. Demolition and Change-Detection Logic

The vendor must implement a first version of demolition/change detection.

7.1 Demolished Candidate Rule

A building should be flagged as `demolished_candidate` if one or more of the following are true:

1. Building appears in older Google/TUM/OSM data but intersects a current road corridor.
2. Building footprint overlaps a newly expanded road surface.
3. Building appears in an older source but is absent or weak in recent temporal raster evidence.
4. Building appears in one global dataset but not in newer source evidence.
5. Building is inside a Tasco-provided demolition or road-construction polygon.
6. Building is visually missing in recent satellite or field imagery.

7.2 Do-Not-Render Rule

A building should be hidden from production rendering if:

```
building_lifecycle_status = demolished_verified
```

A building should be rendered only in QA mode if:


```
building_lifecycle_status = demolished_candidate  
or  
overall_confidence < 0.60  
or  
demolition_confidence >= 0.70
```

7.3 QA Review Rule

The QA dashboard must show all buildings where:

```
building exists in Google/TUM/OSM  
AND building intersects road corridor  
AND building has no current verification
```

This is the most important review queue for the Hanoi pilot.

8. Confidence Scoring Requirements

The vendor must implement separate confidence scores.

8.1 Footprint Confidence

Measures whether the building footprint is geometrically reliable.

Inputs:

```
source agreement  
IoU between OSM/Google/TUM polygons  
centroid distance  
area ratio  
polygon validity  
manual verification  
source recency
```

8.2 Height Confidence

Measures whether the building height is reliable.

Inputs:

```
TUM/GlobalBuildingAtlas height
OSM height
OSM building:levels
raster height prior
local outlier check
manual verification
```

8.3 Freshness Confidence

Measures whether the building is likely to reflect current ground reality.

Inputs:

```
latest source date
temporal raster trend
recent imagery evidence
field survey evidence
road corridor conflict
demolition flag
```

8.4 Demolition Confidence

Measures whether the building is likely demolished or obsolete.

Inputs:

```
intersection with current road corridor
absence in recent imagery or temporal raster
conflict with new road geometry
manual demolition evidence
source age
building-source disagreement
```

8.5 Overall Confidence

The overall confidence should combine:

```
footprint_confidence
height_confidence
freshness_confidence
demolition_confidence
manual QA status
```

9. Required APIs

9.1 Vector Tile API

```
GET /v1/tiles/buildings/{version}/{z}/{x}/{y}.mvt
```

Required attributes:

```
{
  "tasco_building_id": "tasco_bld_hanoi_0000001",
  "height_m": 18.5,
  "legacy_district": "Dong Da",
  "current_ward": "example_ward",
  "footprint_confidence": 0.86,
  "height_confidence": 0.71,
  "freshness_confidence": 0.64,
  "demolition_confidence": 0.22,
  "overall_confidence": 0.78,
  "building_lifecycle_status": "active_building",
  "render_policy": "render_production"
}
```

9.2 3D Tiles API

```
GET /v1/3dtiles/buildings/{version}/hanoi/tileset.json
GET /v1/3dtiles/buildings/{version}/hanoi/{tile_id}.glb
GET /v1/3dtiles/buildings/{version}/hanoi/{tile_id}.b3dm
```

9.3 Building Metadata API

```
GET /v1/buildings/{tasco_building_id}
```

9.4 QA APIs

```
GET /v1/qa/hanoi/demolished-candidates
GET /v1/qa/hanoi/road-corridor-conflicts
GET /v1/qa/hanoi/source-disagreement
GET /v1/qa/hanoi/low-confidence-buildings
GET /v1/qa/hanoi/height-outliers
GET /v1/qa/hanoi/new-construction-candidates
```

9.5 Version and Diff APIs

```
GET /v1/building-versions
```

```
GET /v1/building-versions/{version_id}
```

```
GET /v1/building-versions/{version_id}/diff/{previous_version_id}
```

10. QA Dashboard Requirements

The QA dashboard is mandatory.

10.1 Required Layers

```
canonical buildings
OSM buildings
Google building source
TUM/GlobalBuildingAtlas buildings
current road network
road construction / demolition corridor
demolished candidates
low-confidence buildings
height outliers
source disagreement
do-not-render buildings
```

10.2 Required Filters

```
legacy district / Quận
current ward
building lifecycle status
render policy
footprint confidence
height confidence
freshness confidence
demolition confidence
source combination
manual QA status
```

10.3 Required QA Queues

```
demolished_candidate_queue
road_corridor_conflict_queue
```

```
source_disagreement_queue  
height_outlier_queue  
missing_building_queue  
new_construction_candidate_queue  
manual_review_queue
```

10.4 Required Building Inspector

Clicking on one building should show:

```
tasco_building_id  
geometry source  
height source  
all source lineage  
confidence scores  
lifecycle status  
render policy  
QA flags  
before/after source evidence  
road corridor intersection status  
manual review notes
```

11. MVP Deliverables

The vendor must deliver the following.

11.1 Data Pipeline

```
raw source import scripts  
source snapshot storage  
CRS normalization  
building conflation logic  
canonical building ID generation  
confidence scoring  
demolition/change detection rules  
versioned output
```

11.2 Database

```
PostgreSQL/PostGIS schema  
canonical building table  
source observation table
```

source lineage table
QA ledger table
tile manifest table
version manifest table

11.3 Tile Outputs

MVT vector tiles with height/confidence/status attributes
LoD1 3D building geometry
3D Tiles tileset.json
GLB/B3DM tile payloads

11.4 APIs

vector tile API
3D Tiles API
building metadata API
QA API
version API
diff API

11.5 Web Applications

3D viewer using Cesium or TerriaJS
2D/2.5D viewer using MapLibre
QA dashboard
building inspector
source comparison mode
demolition candidate mode

11.6 Documentation

system architecture document
database schema document
API documentation
data source documentation
confidence scoring documentation
deployment guide
runbook
known limitations
license and source-use notes

12. Acceptance Criteria

The MVP is accepted only if the following are true.

12.1 Data Acceptance

1. The system ingests at least three building data sources.
2. The system creates stable Tasco building IDs.
3. Every rendered building has source lineage.
4. Every rendered building has height_m or explicit height_unknown status.
5. Every rendered building has footprint, height, freshness, demolition, and overall confidence scores.
6. Buildings conflicting with road corridors are flagged.
7. Demolished candidates can be hidden from production rendering.

12.2 Rendering Acceptance

1. The selected Hanoi AOI renders in a web 3D viewer.
2. The same AOI renders through vector tile extrusion.
3. 3D Tiles load correctly through tileset.json.
4. Low-confidence and demolished-candidate buildings can be visually distinguished.
5. Production mode excludes do_not_render buildings.

12.3 QA Acceptance

1. QA dashboard shows demolished candidates.
2. QA dashboard shows road-corridor conflicts.
3. QA dashboard shows source disagreement between OSM, Google, and TUM/Munich data.
4. QA dashboard supports district/ward filtering.
5. Building inspector shows evidence and source lineage.
6. QA status can be updated and stored.

12.4 Reproducibility Acceptance

1. Tasco can rerun the pipeline from raw input sources.
2. Tasco receives source code.
3. Tasco receives Docker or equivalent deployment package.
4. Tasco receives database migration scripts.
5. Tasco receives tile generation scripts.

6. Tasco receives API documentation.
7. Tasco can deploy without relying on vendor-hosted infrastructure.

13. First Technical Spike

Before a full district-scale contract, the vendor should complete a technical spike.

13.1 Spike Duration

10 working days.

13.2 Spike AOI

A 5-10 km² Hanoi urban sample selected by Tasco.

The AOI should include at least one road corridor where building demolition or road expansion is likely to create conflict with older building datasets.

13.3 Spike Inputs

- OSM buildings
- Google or Google-derived building data
- TUM/GlobalBuildingAtlas data
- current road network
- one temporal raster source if available
- one Tasco-provided road corridor or demolition polygon if available

13.4 Spike Outputs

- PostGIS database
- canonical building table
- source lineage table
- basic confidence scoring
- demolished candidate detection
- MVT vector tile output
- 3D Tiles output
- simple 3D viewer
- simple QA dashboard
- short technical report

13.5 Spike Pass/Fail Criteria

PASS if:

- buildings can be generated from multiple sources
- stale/demolished candidates can be flagged
- 3D Tiles can be served and rendered
- vector tiles can be served and rendered
- source lineage is visible per building
- Tasco can rerun the pipeline

FAIL if:

- the vendor only delivers a static 3D visualization
- there is no source lineage
- there is no demolition/change detection
- there is no confidence scoring
- there is no repeatable pipeline
- there is no code handover

14. Ownership and Handover

Tasco must own:

```
source code
database schema
canonical building IDs
confidence scoring logic
tile generation pipeline
QA dashboard
API documentation
deployment scripts
output datasets created for Tasco
```

Vendor-hosted demos are acceptable for presentation, but the production architecture must be deployable in Tasco-controlled infrastructure.

15. Key Questions for Vendor

Please answer the following before project kickoff.

1. Have you generated OGC 3D Tiles from building footprints before?
2. Can you deliver both 3D Tiles and MVT vector tile extrusion?

3. Can you build and document a PostGIS-based canonical building database?
4. Can you preserve source lineage per building?
5. Can you implement building conflation across OSM, Google, and TUM/Munich sources?
6. Can you detect buildings that are obsolete because of road construction or demolition?
7. Can you create separate footprint, height, freshness, demolition, and overall confidence scores?
8. Can you build a QA dashboard for source disagreement and demolition candidates?
9. Can you support both legacy district / Quận AOI segmentation and current ward-level segmentation?
10. Can Tasco rerun the pipeline independently after handover?

16. Final Expected Outcome

The expected outcome is a reproducible Hanoi 3D Building Data and Tile API pilot.

The pilot must not simply show a beautiful 3D city model. It must create a reliable building data product that can determine which buildings should be rendered, which buildings are likely obsolete, which buildings need QA, and which buildings are trusted enough for production map display.

The final MVP must support:

Hanoi district-focused AOI
multi-source building ingestion
canonical building IDs
source lineage
building lifecycle classification
demolition candidate detection
confidence scoring
LoD1 3D building generation
MVT vector tile API
3D Tiles API
QA dashboard
repeatable deployment
Tasco-owned source code and documentation